AI / ML PORTFOLIO ARTIFACT 02

Training a large language model

How data becomes a deployed system, and where resources go

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01 FROM DATA TO DEPLOYMENT

01 Collect data
Gather text and code with documented rights and provenance.
Coverage and licensing shape quality, risk, and acquisition cost.

02 Prepare and tokenize
Deduplicate, filter sensitive content, split held-out data, then tokenize.
Tokens are text units. Clean splits help prevent test leakage.

03 Design and pilot
Choose model size and architecture. Test small runs and a compute budget.
Balance data, memory, and model capacity before scaling up.

04 Pretrain
Predict the next token, measure loss, backpropagate, and update weights.
Repeat across batches on many GPUs. Save recovery checkpoints.

05 Post-train
Teach instructions with demonstrations, then optimize preferences or rewards.
SFT and methods such as RLHF use human or checked synthetic feedback.

06 Evaluate and improve
Test task accuracy, factuality, bias, privacy, and misuse resistance.
Use separate tests. Failed checks can send work back to earlier stages.

07 Deploy and monitor
Optimize and serve the model. Track latency, quality, failures, and use.
Inference needs ongoing compute. Ordinary prompts do not update weights.

Typical workflow. Stages overlap and repeat. Sources [1-3, 6, 8].

02 THE RESOURCE STACK

01 Data
Text, code, storage, curation
Relevant, diverse examples support coverage. Licensing, cleaning, and annotation cost money. More raw data alone does not assure quality.

02 Compute
GPUs or TPUs, memory, networks
Accelerators perform training math. Fast interconnects share work. Memory holds weights, activations, and optimizer state.

03 Energy
Electricity and cooling
Power sustains the cluster. Hardware efficiency and utilization affect use. Carbon also depends on the electricity source and accounting method.

04 Time and people
Cluster hours and project effort
Parallel hardware can shorten a run. Data preparation, experiments, feedback, and evaluation extend the project and require skilled people.

03 WHAT DEVELOPERS HAVE DISCLOSED

GPU-hours shown on a common scale (millions). Hardware and scope differ.
These bars show reported usage, not comparable speed, efficiency, or model quality.

Llama 2 70B
2.0T pretraining tokens
A100 80GB / Pretraining
1.720M GPU-hours [4]
688 MWh power-capacity proxy
Dollar training budget not stated in the cited card.

Llama 3.1 405B
15T+ pretraining tokens
H100 80GB / Training, as reported
30.840M GPU-hours [5]
21.59 GWh power-capacity proxy
Dollar training budget not stated in the cited card.

DeepSeek-V3
14.8T pretraining tokens
H800 / Official training total
2.788M GPU-hours [6]
$5.576M at assumed $2/GPU-hour
Excludes prior research and ablations. Not a total project budget.

GPT-4 (2023): training hardware, compute, model size, and dataset construction
are withheld in its technical report. A verified full training dollar budget is not provided there. [7]

04 READ THE UNITS BEFORE THE PRICE TAG

Time is not the same as GPU-hours
DeepSeek-V3 pretraining: 2.664M GPU-hours [6]
2,664,000 / (2,048 GPUs x 24) = 54.2 days
Calculated at full occupancy. The report says less than two months. Research and preparation add project time.

Energy depends on the boundary
Llama 3.1 405B: 30.84M hours x 0.700 kW [5]
= 21.588M kWh = 21.59 GWh
Capacity-based proxy, not metered electricity. Cooling and other equipment depend on the accounting boundary.

B = billion parameters. T = trillion tokens. A token is a word, word part, or character. GPU-hour = one GPU for one hour.

Source key: [1] Wolfram [2] IBM [3] Ouyang et al. [4] Llama 2 [5] Llama 3.1 [6] DeepSeek-V3 [7] GPT-4 [8] Hoffmann et al.

Full sources, accessible text, calculations, and explanatory document